Mantle cell lymphoma

CD5+, cyclin D1+ B-cell lymphoma driven by t(11;14) · a spectrum from observable leukemic non-nodal disease to blastoid disease · **TP53 status, Ki-67 and fitness — not stage — choose the therapy**; BTK inhibitors have displaced routine ASCT

Compiled from the Stanford Oncology Handbook 2025, Lymphoma: Non-Hodgkin — mantle cell (Chou L, Schroers-Martin J, Gupta NK) · NCCN B-Cell Lymphomas v4.2026 · primary trial reports and FDA actions verified against PubMed, September 2026.

DIAGNOSIS

- **Hallmark: t(11;14)(q13;q32), IGH::CCND1 → cyclin D1 overexpression** — show it by cyclin D1 IHC or FISH
- **Phenotype: CD5+, CD20 bright, CD23–, CD10–, cyclin D1+, SOX11+.** CD23 negativity and cyclin D1 separate it from CLL, which is CD5+ CD23+
- **Cyclin D1-negative MCL is real:** SOX11+, with a **CCND2 rearrangement in ~55%**. Diagnose it on SOX11 + morphology; treat as MCL *Salaverria, Blood 2013*
- **SOX11 does double duty** — rescues the cyclin D1-negative diagnosis, and its **absence flags the indolent leukemic non-nodal form**

THE SPECTRUM — WHO-HAEM5 / ICC 2022

Form	Features → behavior
Classic (nodal)	SOX11+, IGHV mostly unmutated; nodes + marrow, spleen, **GI tract** (lymphomatous polyposis). Chronic relapsing course
Leukemic non-nodal	**SOX11–, IGHV-mutated**, blood + marrow ± spleen, no significant adenopathy. **Indolent**: 5-yr OS **78% vs 36%** for SOX11+ *Fernàndez 2010* — **observe** unless it acquires TP53 or goes nodal
Blastoid / pleomorphic	High Ki-67, TP53 enriched, CNS-prone. **Treat as high risk**; blastoid adds nothing to prognosis once Ki-67 is known
In situ mantle cell neoplasia	Cyclin D1+ cells confined to mantle zones of a reactive node — usually incidental; stage and observe

WORKUP

- **PET/CT**, marrow biopsy with flow and t(11;14) FISH, CBC with smear (leukemic phase), LDH, β2-microglobulin, HBV serology before anti-CD20
- **TP53 by NGS (+ del(17p) FISH) and Ki-67 at diagnosis in every patient** — they pick the regimen, not the stage
- **EGD + colonoscopy** for apparent stage I–II (to confirm limited disease) and when GI symptoms
- **LP / brain MRI** if blastoid or neurologic symptoms. Echo before anthracycline

PEARLS

- **CD5+ CD23– small B cells = send cyclin D1.** If cyclin D1– but SOX11+, it is still MCL
- **Send TP53 before choosing.** A TP53-mutant patient gets little from cytarabine + ASCT
- **Rituximab maintenance is standard in MCL** after both intensive and BR induction — unlike follicular, LyMa showed an OS gain at 4 yr (attenuated by 7)
- **No BTKi-added frontline trial in older patients has shown OS gain** (SHINE, ECHO) — PFS gain costs toxicity
- **Brexu-cel is the most active post-BTKi therapy** but the most neurotoxic; liso-cel trades a little efficacy for much less CRS/ICANS

MIPI

Pts	Age	ECOG	LDH / ULN	WBC $\times 10^9$/L
0	<50	0–1	<0.67	<6.7
1	50–59	—	0.67–0.99	6.7–9.99
2	60–69	2–4	1.00–1.49	10–14.99
3	≥70	—	≥1.50	≥15

Low 0–3 (44%; median OS not reached) · **intermediate 4–5** (35%; 51 mo) · **high 6–11** (21%; 29 mo). Hoster, Blood 2008 — pre-BTKi survival; use it to stratify, not to counsel.

HIGH-RISK BIOLOGY

- **TP53 mutation is the dominant adverse factor.** *11% in Nordic MCL2/3* Median OS **1.8 vs 12.7 yr**, OS HR 6.2; half relapsed within a year of cytarabine induction + ASCT *Eskelund, Blood 2017*
- **Ki-67 ≥30%** is the validated cutoff. **MIPI-c** = MIPI group + Ki-67 → 4 groups, 5-yr OS **85%** / **72%** / **43%** / **17%** *Hoster, JCO 2016*
- **Blastoid/pleomorphic cytology, complex karyotype, del(17p)** travel with TP53 and high Ki-67

> **Watch and wait is legitimate for selected patients** — leukemic non-nodal SOX11– disease, or asymptomatic classic MCL with low burden, low Ki-67 and wild-type TP53. In a Weill Cornell series **32% were observed**; median time to treatment **12 mo (range 4–128)**, with no survival penalty (a selected cohort) *Martin, JCO 2009*.

> **ASCT in first remission is no longer default.** Its PFS benefit predates BTKi and maintenance. **TRIANGLE**: adding ASCT to ibrutinib gave no FFS gain and more toxicity. **EA4151**: in MRD-negative CR, ASCT + rituximab = rituximab alone (abstract only).

CHEMO-IMMUNOTHERAPY — INFECTION AND SUPPORTIVE CARE

- **Bendamustine is deeply immunosuppressive** — prolonged CD4 lymphopenia: **PJP (TMP-SMX) + VZV (acyclovir) prophylaxis**; ECHO grade 5 COVID-19 **9.4% vs 6.7%**. Rash with allopurinol
- **Rituximab maintenance** → hypogammaglobulinemia and infection; check IgG, replace if recurrent infections. HBV prophylaxis if core Ab+
- **Lenalidomide:** aspirin or anticoagulant VTE prophylaxis, renal dose adjustment

TREATMENT PATHWAY

Frontline

Indolent / low burden, TP53 wild-type
leukemic non-nodal, or low Ki-67 asymptomatic
observe — treat for symptoms or progression

Fit, ≤65–70, TP53 wild-type
TRIANGLE: R-CHOP + ibrutinib alternating with R-DHAP (or R-DHAOx) ×6 → **ibrutinib ×2 yr**
± rituximab q8wk — **ASCT can be omitted**
older standard: Nordic or R-DHAP → ASCT → R q2mo ×3 yr

Older or unfit, TP53 wild-type
acalabrutinib + BR (ECHO) → R maintenance
BR → R maintenance · VR-CAP · RBAC500 (fit)
chemo-free: BTKi + rituximab (ENRICH)

TP53-mutated
avoid chemo-heavy regimens and ASCT
clinical trial, or BTKi-based: BOVen
(zanubrutinib + obinutuzumab + venetoclax)

Relapsed / refractory

BTKi-naive
covalent BTKi: acalabrutinib or zanubrutinib
or ibrutinib + venetoclax (SYMPATICO, off-label)

After a covalent BTKi
CAR-T: brexu-cel or liso-cel (fit)
pirtobrutinib — bridge, or if CAR-T not possible
sonrotoclax · R^2 · glofitamab on trial

After CAR-T
allo-HCT if fit · trials (bispecifics, BTK degraders)

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FRONTLINE EVIDENCE

Trial	Result	How to read it
TRIANGLE *≤65, n=870*	Ibrutinib + ASCT (A+I) beat ASCT alone (A): 3-yr FFS **88% vs 72%**, HR 0.52. At 4.5 yr **A+I not better than ibrutinib without ASCT (I)**: 4-yr FFS 82% vs 81%. **4-yr OS 88–90% vs 81%** for either ibrutinib arm vs A (HR 0.57–0.59)	Primary endpoint is **FFS**. Maintenance grade 3–5 heme toxicity **54% A+I vs 28% I**. Uses ibrutinib, which has **no US MCL label**; applying it to acalabrutinib/ zanubrutinib is extrapolation
TRIANGLE R-maint *secondary*	Rituximab maintenance on top of ibrutinib: 4-yr PFS **85% vs 73%** (arm I)	Non-randomized; grade 3–5 infections **34% vs 11%**
LyMa *<66, n=240 rand.*	R-DHAP → ASCT → **rituximab q2mo ×3 yr** vs observation: 4-yr EFS **79% vs 61%**, OS **89% vs 80%** (HR 0.50)	PFS gain holds at 7 yr (HR 0.36); **OS gain no longer significant**
EA4151 *MRD– CR1, n=516 rand.*	ASCT + R maintenance vs **R maintenance alone**: 3-yr OS **82.1% vs 82.7%** (HR 1.11); PFS 76.6% vs 77.4%. **Crossed futility**	**Abstract only**; few had BTKi induction
Nordic MCL2 *single-arm*	Maxi-R-CHOP / R-high-dose cytarabine → ASCT: median PFS **8.5 yr**, OS **12.7 yr**	No plateau — not curative
SHINE *≥65, n=523*	Ibrutinib + BR → R maintenance: median PFS **80.6 vs 52.9 mo** (HR 0.75)	**No OS benefit** — excess adverse-event deaths; AF **13.9% vs 6.5%**
ECHO *≥65, n=598*	Acalabrutinib + BR: median PFS **66.4 vs 49.6 mo** (HR 0.73). **Regular FDA approval, Jan 2025**	**No OS difference** (HR 0.86). AF 6.7% vs 4.4%; grade 5 COVID-19 9.4% vs 6.7%
ENRICH *≥60, n=397*	Ibrutinib + rituximab vs R-chemo: PFS HR **0.69**. vs R-CHOP HR 0.37; **vs BR HR 0.91 (NS)**	Chemo-free BTKi-R **matches BR** and beats R-CHOP
VR-CAP *n=487*	Bortezomib for vincristine in R-CHOP: PFS **24.7 vs 14.4 mo**; median OS **90.7 vs 55.7 mo** (HR 0.66)	The one older-patient trial with an OS gain
RBAC500 *single-arm, n=57*	R + bendamustine + cytarabine 500: **CR 91%**, median age 71	Missed its safety boundary: **72% dose-reduced**, grade 3–4 cytopenias in ~half of cycles
BOVen *TP53-mut, n=25*	Zanubrutinib + obinutuzumab + venetoclax: ORR **96%**, CR **88%**, 2-yr PFS **72%**	Single-arm phase 2 — the best TP53 data, not yet randomized

RELAPSED / REFRACTORY — AGENTS AND STATUS (SEPT 2026)

Agent	Data	FDA status
Acalabrutinib	ACE-LY-004 (n=124): ORR **81%**, CR **40%**	Accelerated 2017 → **regular Jan 2025**
Zanubrutinib	BGB-3111-206 (n=86): ORR **84%**, CR 69%, median PFS 22.1 mo	**Accelerated** (Nov 2019), still
Ibrutinib	Pivotal single-arm data; SHINE did not confirm	**US MCL indication withdrawn** (effective Dec 2023)
Ibrutinib + venetoclax	SYMPATICO (n=267): median PFS **31.9 vs 22.1 mo**, HR 0.65 vs ibrutinib	**Not approved** as a combination in MCL
Pirtobrutinib *non-covalent*	BRUIN, after covalent BTKi (n=90): ORR **58%**, CR 20%, median DOR 21.6 mo; grade ≥3 AF 1.2%	**Accelerated** (Jan 2023; ≥2 lines incl. BTKi). Confirmatory **BRUIN MCL-321 not yet reported**
Brexu-cel *CD19 CAR-T*	ZUMA-2: ORR **93%**, CR **67%**; 3-yr median PFS **25.8 mo**, OS 46.6 mo. Grade ≥3 CRS 15%, **neurologic events 31%**	Accelerated 2020 → **regular Apr 2026** (added BTKi-naive cohort: CR 79%)
Liso-cel *CD19 CAR-T*	TRANSCEND MCL (n=83): ORR **83%**, CR **72%**, median PFS 15.3 mo. Grade 3–4 CRS **1%**, neurologic 9%	Approved May 2024 (≥2 lines incl. BTKi)
Sonrotoclax *BCL2*	Phase 2 (n=103), post-BTKi: ORR **52%**, CR 16%, median PFS **6.5 mo**; TP53-mut ORR 59%; TLS 7%	**Accelerated, 13 May 2026** (≥2 lines incl. BTKi)
Glofitamab *CD20×CD3*	Phase 1/2 (n=60): ORR **85%**, CR **78%** (71% after BTKi); CRS 70%	**Not approved in MCL** (US or EU) — phase 3 GLOBRYTE pending
Lenalidomide ± R	SPRINT: PFS **8.7 vs 5.2 mo** vs investigator choice (HR 0.61); EMERGE post-bortezomib ORR 28%	Approved 2013 (after bortezomib)
Allo-HCT	Only curative-intent option after CAR-T failure	—

Relapse after frontline BTKi (TRIANGLE, ECHO) is now common: those patients are BTKi-exposed at first relapse, so the post-BTKi column is effectively second line. Response to a covalent BTKi at relapse after BTKi-containing induction is not established.

REGIMEN DOSING

Regimen	Dosing
TRIANGLE (A+I / I)	**R-CHOP + ibrutinib**: rituximab 375, cyclophosphamide 750, doxorubicin 50, vincristine 1.4 (max 2 mg) d1, prednisone 100 mg d1–5 + **ibrutinib 560 mg d1–19**, alternating with **R-DHAP** (rituximab 375, dexamethasone 40 mg d1–4, **cytarabine 2 g/m² q12h ×2 d2**, cisplatin 100 CIV d1) or R-DHAOx (oxaliplatin 130) — q21d ×6. Then **ibrutinib 560 mg daily ×2 yr** ± rituximab
LyMa	R-DHAP q21d ×4 → ASCT → **rituximab 375 q2 months ×3 yr**
BR ± BTKi	Rituximab 375 d1 + **bendamustine 90 d1–2**, q28d ×6 → rituximab 375 q8wk ×2 yr. ECHO adds **acalabrutinib 100 mg BID** until progression
VR-CAP	Rituximab 375, cyclophosphamide 750, doxorubicin 50 d1; **bortezomib 1.3 d1, 4, 8, 11**; prednisone 100 d1–5 — q21d ×6
RBAC500	Rituximab 375 d1, **bendamustine 70 d2–3**, **cytarabine 500 d2–4** — q28d, up to 6
R²	Lenalidomide 20 mg d1–21 q28d + rituximab 375 weekly ×4 (cycle 1), then per protocol. VTE prophylaxis
BTK inhibitors	Ibrutinib 560 mg daily · **acalabrutinib 100 mg BID** · **zanubrutinib 160 mg BID or 320 mg daily** · **pirtobrutinib 200 mg daily** — continuous
CAR-T	**Brexu-cel**: flu 30 + cy 500 mg/m² d–5 to –3, then 2×10^6 CAR-T cells/kg (max 2×10^8). **Liso-cel**: flu 30 + cy 300 mg/m² ×3 d, then $90–110\times10^6$ CAR+ T cells. **Avoid bendamustine shortly before apheresis**

mg/m² unless stated. Cytarabine: **prednisolone 1% eye drops** from 12 h before to 48 h after; **reduce to 1 g/m² if age >60**; hold for rising creatinine (cerebellar toxicity).

TARGETED AND CELLULAR THERAPY — TOXICITY

- **BTKi class: atrial fibrillation, bleeding, hypertension, infection**, plus arthralgia, diarrhea, rash. Hold **3–7 days around procedures**; avoid warfarin; watch CYP3A interactions
- **Second-generation agents cut AF by about half:** ELEVATE-RR (CLL) **9.4% acalabrutinib vs 16.0% ibrutinib**; ALPINE fewer cardiac events with zanubrutinib; in MCL, ECHO 6.7% vs SHINE 13.9% (cross-trial). Acalabrutinib: early headache. Zanubrutinib: neutropenia. Pirtobrutinib: grade ≥3 AF ~1%
- **CAR-T:** CRS → tocilizumab ± dexamethasone; **ICANS** → dexamethasone (tocilizumab only if CRS coexists), levetiracetam. **Brexu-cel is the more neurotoxic** (grade ≥3 neuro 31% vs 9% liso-cel). Late: cytopenias, hypogammaglobulinemia, boxed warning for T-cell malignancy
- **Venetoclax / sonrotoclax: TLS** — MCL can be bulky and leukemic; ramp-up, hydration, allopurinol, lab checks at each step